Performance Testing

Date	20 October 2026
Team ID	Individual project (no team)
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	5 Marks

Step 1: Testing Overview

Testing Tool Used	Custom Python load script (perf_test.py using requests and threading)
Type of Testing	Light load testing
Target Module / API	GET / (health), GET /trips (database read), POST /itinerary (LLM generation)
Test Environment	Cloud - backend on Render free tier, database on Supabase free tier, tests run from a personal laptop
Test Date	**[FILL test date]**

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Health check: GET /	10	20	All requests return 200 quickly
2	Read saved trips: GET /trips	10	20	Rows returned with low error rate
3	Generate itinerary: POST /itinerary (Goa, 2 days)	2	40	Valid JSON itinerary returned for every request
4	Cold start: first request after the free tier has been idle	1	-	Response within about 60 seconds

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 s (non-AI); < 20 s (/itinerary)	**[FILL]**	**[FILL]**	From perf_test.py output
2	Response Time (Max)	< 5 s (non-AI); < 40 s (/itinerary)	**[FILL]**	**[FILL]**	
3	Throughput (Req/sec)	-	**[FILL]**	-	Limited by the LLM for /itinerary
4	Error Rate	< 1%	**[FILL]**	**[FILL]**	
5	CPU Utilization	< 80%	**[FILL]**	**[FILL]**	From the Render Metrics tab
6	Memory Utilization	< 80%	**[FILL]**	**[FILL]**	From the Render Metrics tab

Step 4: Observations & Analysis

[FILL: write 3-4 lines after running the test - for example which endpoint was slowest, whether the cold start was noticeable, and whether any errors or rate limits appeared.]

Step 5: Screenshots / Evidence

[FILL: paste a screenshot of the perf_test.py output and of the Render Metrics tab here.]